Combined DR1+DR2 BAO Fit Identifies a New Lock Candidate (LockPhi) at $z_c \approx 1/\varphi$

LVC v16 — Working Paper

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Abstract

The v15 working paper found that DESI DR1 BAO and DESI DR2 BAO, when used as alternative replacements in the same baseline (Pantheon+ unbinned with full STAT+SYS covariance, Planck 2018 compressed distance priors, SH0ES H_0, BOSS DR12, eBOSS DR16), prefer different positions in the locked-Gaussian parameter (z_c, w). DR2 returns the C1 region ($z_c \approx 0.683$) and DR1 returns $(z_c, w) \approx (0.568, 0.127)$, identified in v15 as LockB and LockC.

This v16 working paper reports the outcome of placing DR1 and DR2 in the likelihood simultaneously rather than as alternatives. With BOSS, eBOSS, Pantheon+, the distance priors and SH0ES included once, this gives $N_{\text{combined}} = 1738$. A free six-parameter Gaussian fit to the combined likelihood returns $z_c = 0.6185$, $w = 0.172$. The natural-constant $1/\varphi = 0.61803$ matches the free-fit z_c to 0.075%. Two natural-constant candidates for w are within 1.5%: $e/(5\pi) = 0.17305$ and $\pi/18 = 0.17453$. We define LockPhi as the family $z_c = 1/\varphi$ with $w \in \{e/(5\pi), \pi/18\}$. The two members are not distinguishable in χ^2 on the present data ($\Delta\chi^2 = 0.002$). For all numerical work below we use the form $w = e/(5\pi)$ (LockPhi-A) unless stated.

The combined-fit ΔBIC values relative to flat ΛCDM at $N = 1738$ are: LockB -12.22, LockC -13.27, LockPhi -16.20. A scan of natural-constant candidates for z_c within $\pm1.5\%$ of $1/\varphi$ ($5/8$, $\sqrt{\pi/8}$, $11/18$, $1/\sqrt{e}$, plus a 0.5%-spaced manual sweep) returns $\Delta\chi^2$ between the best ($1/\varphi$) and the closest non-φ candidate (the round number 0.6200) of $+0.0004$; ten candidates spanning $z_c \in [0.607, 0.630]$ all lie within $\Delta\chi^2 < 0.14$ of the minimum. A two-dimensional χ^2 landscape on (z_c, w) within $\pm5\%$ of z_c and $\pm15\%$ of w returns 21 of 25 grid points with $\Delta\chi^2 < 1.0$ from the LockPhi center; all 25 grid points satisfy $\Delta\text{BIC} < -14$ relative to flat ΛCDM. We therefore record LockPhi as a candidate at the center of a ridge, with the natural-constant identification of z_c being a structural fact about where the data points and not a parameter saving in the BIC sense (working principle (e), as in v15 §5).

Direct kill tests are run on the LockPhi center. K-A multi-LOO (eleven combinations of subset removal of {LRG1, LRG2, BOSS, eBOSS_LRG} from both BAO releases simultaneously) returns 7 WIN, 4 tie, 0 BURN at $\Delta\text{BIC} = -2$ threshold; the worst case (LRG1+LRG2+BOSS+eBOSS_LRG removed, total 12 measurements dropped) gives $\Delta\text{BIC} = +0.66$ tie. The free-Ω_Λ test returns LockPhi $\Delta\text{BIC} = -22.72$ vs free-Ω_Λ ΛCDM (kCDM); the lock amplitude A shifts by $+17\%$ from the flat-Ω_Λ value. The $f\sigma_8$ joint analysis returns $\chi^2(f\sigma_8) = 10.075$ for LockPhi vs 10.069 for flat ΛCDM, joint $\Delta\text{BIC} = -16.18$. The BBN-strict $r_d = 147.05 \pm 0.3\,\text{Mpc}$ prior returns LockPhi $\Delta\text{BIC} = -10.53$, a shift of $+5.67$ from the unforced value. A point-by-point BAO decomposition assigns approximately 75% of the combined BAO improvement to four measurements (DR1/DR2 LRG1 at $z = 0.510$, DR1/DR2 LRG2 at $z = 0.706$); an A-profile scan with all four removed returns best-fit $A = +0.040$ ($\Delta\chi^2 = -7.73$, $\Delta\text{BIC} = -0.27$ tie).

LockPhi is a new candidate identified by the combined-fit analysis. The v15 status of LockB and LockC as live, not closed is unchanged. We do not advance further candidates. The next decisive test is DESI DR3.

1 Setup

The model class is unchanged from v10 onward: a multiplicative modulation of the ΛCDM expansion rate,

$$H(z) = H_{\Lambda\mathrm{CDM}}(z; H_0, \Omega_m) \cdot T(z), \tag{1}$$

with $T(z) \to 1$ at $z = 0$ and $z \gg 1$, r_d derived geometrically from θ_* as in v14, and modulation amplitude A as the only new parameter beyond ΛCDM in any single locked candidate. The modulation is taken to be a single Gaussian,

$$T(z) = 1 + A \exp\left[-\left(\frac{z - z_c}{w}\right)^2\right]. \tag{2}$$

Different locked candidates fix (z_c, w) to different natural-constant values; only $(H_0, \Omega_m, \Omega_b h^2, A)$ are then free ($k = 4$).

The combined likelihood used in this paper places DR1 BAO and DR2 BAO in the same fit, with the remaining data identical to v15:

- Pantheon+ unbinned: 1701 supernovae with full STAT+SYS covariance and 77 Cepheid-host calibrators.
- DESI DR2 BAO: 1 isotropic D_V at $z = 0.295$ and six anisotropic D_M/r_d, D_H/r_d pairs at $z \in \{0.510, 0.706, 0.934, 1.321, 1.484, 2.330\}$ (13 measurements).
- DESI DR1 BAO: 1 isotropic D_V at $z = 0.295$ and five anisotropic pairs at $z \in \{0.510, 0.706, 0.930, 1.317, 2.33$ (11 measurements).
- BOSS DR12: D_M, D_H at $z = 0.38$ (2 measurements, included once).
- eBOSS DR16: D_M, D_H at $z \in \{0.698, 1.480, 2.334\}$ and D_V at $z = 0.845$ (7 measurements, included once).
- Planck θ_* used to derive r_d in geometric fits.
- SH0ES $H_0 = 73.04 \pm 1.04\,\mathrm{km/s/Mpc}$ (one measurement).
- Planck 2018 compressed distance priors (R, ℓ_A, ω_b) with the multiplicative calibration applied identically to all candidates as in v14.

The total measurement count is $N = 1701 + 13 + 11 + 2 + 7 + 3 + 1 = 1738$. The DR1 BAO and DR2 BAO blocks share three labels (BGS at $z = 0.295$, LRG1 at $z = 0.510$, LRG2 at $z \approx 0.7$) but the measured D_M/r_d, D_H/r_d values differ between releases (precision and central-value updates). Both blocks are included as independent measurements. BOSS and eBOSS appear once.

The Pantheon+ likelihood, distance-prior likelihood and BIC accounting are exactly those of v15. The flat ΛCDM combined fit ($k = 3$: H_0, Ω_m, $\Omega_b h^2$) returns $\chi^2 = 1624.85$. The v15 environment with single-release BAO is reproduced to better than 0.005 in χ^2.

2 Combined fit baseline (ΛCDM)

Flat ΛCDM fit on the combined likelihood:

	χ^2	H_0	Ω_m	$\Omega_b h^2$	r_d
ΛCDM, $k = 3$	1624.85	70.40	0.2764	0.02286	145.59

The χ^2-decomposition by data block at the flat-ΛCDM best fit is: Pantheon+ 1545.77; DR2 BAO 24.91; DR1 BAO 17.80; BOSS 2.29; eBOSS 5.75; distance priors 21.91; SH0ES residual 6.42. The SH0ES residual at $H_0 = 70.40$ is $(73.04 - 70.40)/1.04 = 2.54$ standard deviations.

3 LockB and LockC on combined data

We first apply the two v15 surviving candidates to the combined likelihood. The lock parameters are fixed to the v15 values; only $(H_0, \Omega_m, \Omega_b h^2, A)$ are fit.

	χ^2	H_0	Ω_m	A	$\Delta\chi^2$	ΔBIC
LockB ($z_c = 1/\sqrt{\pi}$, $w = 2/(5\pi)$)	1605.17	69.53	0.2852	+0.0593	−19.68	−12.22
LockC ($z_c = 4/7$, $w = 2/(5\pi)$)	1604.12	69.51	0.2855	+0.0619	−20.73	−13.27

Both are stronger by BIC than their single-release values reported in v15 (DR2: −5.29, −6.21; DR1: −3.60, −3.60). The combined ΔBIC is approximately the sum of the two single-release contributions to $\Delta\chi^2$, minus the single $\ln N$ penalty (rather than two separate penalties).

4 Free-Gaussian combined fit

We open the lock entirely on the combined data. The free-Gaussian model with all six parameters $(H_0, \Omega_m, \Omega_b h^2, A, z_c, w)$ free, fit with five seeds of differential evolution followed by Nelder–Mead polish:

	χ^2	H_0	Ω_m	A	z_c	w
free Gaussian, $k = 6$	1601.19	69.43	0.2864	+0.0522	0.6185	0.1720

All five seeds converge to the same minimum within $\Delta\chi^2 < 0.05$. The combined free best $z_c = 0.6185$ lies between the v15 single-release free best values 0.5681 (DR1) and 0.6691 (DR2); the combined free best $w = 0.1720$ lies between 0.1273 (DR1) and 0.2151 (DR2). The combined free best is closer to the DR1 free best in z_c (relative deviation 8.9% vs 7.6% from DR2) and intermediate in w.

5 LockPhi candidate

5.1 Definition

The combined free-fit $z_c = 0.6185$ admits the natural-constant interpretation

$$\frac{1}{\varphi} = \frac{2}{1+\sqrt{5}} = 0.61803\ldots, \qquad \text{deviation } -0.075\%, \tag{3}$$

where $\varphi = (1+\sqrt{5})/2$ is the golden ratio. The combined free-fit $w = 0.1720$ admits two close interpretations,

$$\frac{e}{5\pi} = 0.17305\ldots\ (-0.6\%), \qquad \frac{\pi}{18} = 0.17453\ldots\ (-1.5\%), \tag{4}$$

which are not distinguishable on the present data: locking $z_c = 1/\varphi$ and refitting at fixed w returns $\chi^2 = 1601.189$ for $w = e/(5\pi)$ and $\chi^2 = 1601.191$ for $w = \pi/18$. We label these LockPhi-A and LockPhi-B and use LockPhi-A as the reference form below; tabulated ΔBIC values for the two forms agree to within 0.005.

The combined-fit ΔBIC table including LockPhi:

	k	χ^2	ΔBIC vs ΛCDM
flat ΛCDM (reference)	3	1624.85	baseline
LockB	4	1605.17	−12.22
LockC	4	1604.12	−13.27
LockPhi-A ($z_c = 1/\varphi$, $w = e/(5\pi)$)	4	1601.19	−16.20
LockPhi-B ($z_c = 1/\varphi$, $w = \pi/18$)	4	1601.19	−16.20
free Gaussian	6	1601.19	— (penalty applies separately)

5.2 Natural-constant scan in z_c

To test whether the data prefer $1/\varphi$ over other natural-constant candidates near the combined free best, we lock $w = e/(5\pi)$ and scan z_c over five natural-constant candidates within 1.5% of 0.6185, plus five round-number fillers at 0.5% spacing for granularity. At each z_c we refit $(H_0, \Omega_m, \Omega_b h^2, A)$.

label	formula	z_c	χ^2	$\Delta\chi^2$ vs $1/\varphi$
$1/\varphi$	golden ratio	0.61803	1601.1894	0.0000
0.6200	round number	0.62000	1601.1897	+0.0004
0.6150	round number	0.61500	1601.2020	+0.0126
5/8	simple fraction	0.62500	1601.2199	+0.0305
$\sqrt{\pi/8}$	natural constant	0.62666	1601.2388	+0.0495
11/18	simple fraction	0.61111	1601.2423	+0.0529
0.6100	round number	0.61000	1601.2589	+0.0695
0.6300	round number	0.63000	1601.2899	+0.1005
$1/\sqrt{e}$	natural constant	0.60653	1601.3254	+0.1360

The minimum is at $1/\varphi$, but the second-best candidate ($z_c = 0.6200$, a round number with no natural-constant interpretation) lies $\Delta\chi^2 = +0.0004$ above. All ten candidates lie within $\Delta\chi^2 < 0.14$ of the minimum. The data do not resolve z_c within the range $[0.607, 0.630]$ at the $\Delta\chi^2 < 1$ level. The lock at $z_c = 1/\varphi$ was identified by inspection of the combined free-fit best; by working principle (e) of the project (as adopted in v15 §5 for LockB/C), this does not save a parameter, and we count $k = 4$ for LockPhi.

5.3 Two-dimensional χ^2 landscape

We extend the previous scan to (z_c, w) jointly. A 5×5 grid is taken with z_c at ±5% around $1/\varphi$ and w at ±15% around $e/(5\pi)$. At each grid point (z_c, w), we fit $(H_0, \Omega_m, \Omega_b h^2, A)$. Entries below are ΔBIC relative to flat ΛCDM.

$z_c \backslash w$	0.1471	0.1601	0.1731	0.1860	0.1990
0.5871	−15.34	−15.33	−15.27	−15.18	−15.07
0.6026	−16.09	−16.04	−15.96	−15.86	−15.74
0.6180 ($1/\varphi$)	−16.12	−16.18	**−16.20**	−16.17	−16.09
0.6335	−15.53	−15.83	−16.03	−16.12	−16.12
0.6489	−14.55	−15.13	−15.53	−15.77	−15.87

The minimum of the scanned grid coincides with the LockPhi center $(1/\varphi, e/(5\pi))$ at $\Delta\text{BIC} = -16.20$. Twenty-one of twenty-five grid points lie within $\Delta\chi^2 < 1.0$ of this minimum. All twenty-five grid points have $\Delta\text{BIC} < -14.5$ relative to flat ΛCDM. The best-fit amplitude across the grid stays in $A \in [0.046, 0.059]$ with no sign change. The combined-fit minimum is real (the LockPhi center is the global minimum within the scanned region) but the surrounding ridge is wide.

6 Direct kill tests

6.1 K-A multi-leave-one-out on combined BAO

We extend the v15 single-release K-A sweep (v15 appendix Tables 2a/2b) to the combined likelihood. Eleven subsets of {LRG1, LRG2, BOSS, eBOSS_LRG} are removed simultaneously from both DR1 and DR2 BAO blocks (the same label appearing in both releases is removed in both). $(H_0, \Omega_m, \Omega_b h^2, A)$ are refit at each subset removal at fixed $(z_c, w) = (1/\varphi, e/(5\pi))$.

subset removed	ΔN	$\Delta\chi^2$	ΔBIC	verdict
eBOSS_LRG	2	−23.18	−15.72	WIN
BOSS	2	−22.14	−14.68	WIN
LRG1	4	−17.48	−10.02	WIN
LRG1+eBOSS_LRG	6	−17.14	−9.68	WIN
LRG1+BOSS	6	−16.00	−8.54	WIN
LRG1+BOSS+eBOSS_LRG	8	−15.65	−8.20	WIN
LRG2	4	−13.68	−6.22	WIN
LRG1+LRG2	8	−7.74	−0.28	tie
LRG1+LRG2+eBOSS_LRG	10	−7.25	+0.20	tie
LRG1+LRG2+BOSS	10	−7.21	+0.24	tie
LRG1+LRG2+BOSS+eBOSS_LRG	12	−6.80	+0.66	tie

Verdict count: 7 WIN, 4 tie, 0 BURN at the $\Delta\text{BIC} = -2$ threshold. The DR1 LRG1 single-LOO failure of v15 §5 (LockB/C $\Delta\text{BIC} = +1.87$) is replaced in the combined likelihood by LRG1 simultaneous removal from both releases, which gives $\Delta\text{BIC} = -10.02$ (WIN). The four tie entries all involve simultaneous removal of LRG1 and LRG2.

6.2 Free-Ω_Λ test

We replace the flatness assumption $\Omega_\Lambda = 1 - \Omega_m - \Omega_r$ with Ω_Λ free, allowing curvature $\Omega_k = 1 - \Omega_m - \Omega_r - \Omega_\Lambda$. This adds one parameter to the LCDM model (now kCDM, $k = 4$) and one to the lock model ($k = 5$).

	k	χ^2	H_0	Ω_m	Ω_Λ	A	ΔBIC*
flat ΛCDM (reference)	3	1624.85	70.40	0.2764	0.7236	—	baseline
flat LockPhi	4	1601.19	69.43	0.2864	0.7136	+0.0520	−16.20
free-Ω_Λ ΛCDM (kCDM)	4	1619.80	71.11	0.2751	0.7432	—	+2.41
free-Ω_Λ LockPhi	5	1589.62	70.36	0.2858	0.7438	+0.0610	−20.31

* all ΔBIC values are vs flat ΛCDM with appropriate parameter penalty.

The lock amplitude under free Ω_Λ is $A = +0.0610$, a +17.1% shift from the flat value $A = +0.0520$. Same sign, same order of magnitude. The free-Ω_Λ LCDM solution prefers a slightly closed universe ($\Omega_k = -0.018$) and improves on flat LCDM by $\Delta\chi^2 = -5.05$, but does not pass BIC ($\Delta\text{BIC} = +2.41$). The LockPhi vs kCDM ΔBIC is −22.72: LockPhi remains preferred over ΛCDM after ΛCDM is given the curvature degree of freedom.

6.3 $f\sigma_8$ joint analysis

The 19-point $f\sigma_8$ compilation of v11 §3 is added to the combined likelihood, with linear growth $D(z)$ and growth rate $f(z)$ obtained by integrating the standard scale-independent linear-growth ODE with $H(z) = H_{\Lambda\text{CDM}}(z) \cdot T(z)$ and initial conditions $D = a$, $dD/d\ln a = a$ at $a_{\text{init}} = 10^{-3}$. The amplitude $\sigma_8(0)$ is profiled analytically. Background parameters are taken from each model's combined-fit best.

	Ω_m	$\sigma_8(0)$	$\chi^2(f\sigma_8)$	$\Delta\chi^2$ vs ΛCDM	joint ΔBIC
ΛCDM	0.2764	0.8144	10.07	—	baseline
LockB	0.2852	0.8230	9.92	−0.15	−12.36
LockC	0.2855	0.8236	9.97	−0.10	−13.36
LockPhi	0.2864	0.8247	10.08	+0.006	−16.18

The LockPhi $\chi^2(f\sigma_8)$ matches ΛCDM to within 0.006. The joint geometry $+f\sigma_8$ ΔBIC at $N_{\text{joint}} = 1738 + 19 = 1757$ shifts from the geometry-only −16.20 to −16.18 (a change of +0.02).

6.4 BBN-strict r_d prior

A Gaussian prior $r_d = 147.05 \pm 0.3\,\mathrm{Mpc}$ is added to enforce BBN-compatible early-universe physics. This is the v15 §5 BBN test on the combined likelihood. Both ΛCDM and LockPhi are refit.

	χ^2 (incl prior)	r_d (Mpc)	A	ΔBIC vs ΛCDM
ΛCDM (BBN-forced)	1642.82	145.95	—	baseline
LockPhi (BBN-forced)	1624.83	145.80	+0.0449	−10.53

Reference shifts from v15 §5 (LockC, single-release): DR2 −6.21 → −2.46 (shift +3.75); DR1 −3.60 → −1.75 (shift +1.85). LockPhi (combined): −16.20 → −10.53 (shift +5.67). ΛCDM also fails to reach $r_d = 147.05$ at the combined best fit ($r_d = 145.95$); the BBN-forced LCDM prior contribution is 13.56. The lock amplitude under BBN-strict is $A = +0.0449$, a −13.7% shift from the unforced value +0.0520. Same sign, similar magnitude.

6.5 χ^2 decomposition and BAO point-by-point

At the LockPhi combined best, the χ^2 contribution by data block:

block	ΛCDM χ^2	LockPhi χ^2	$\Delta\chi^2$
Pantheon+ (1701 SN)	1545.77	1543.69	−2.08
DESI DR2 BAO (13)	24.91	12.13	−12.77
DESI DR1 BAO (11)	17.80	12.02	−5.78
BOSS DR12 (2)	2.29	0.92	−1.37
eBOSS DR16 (7)	5.75	5.01	−0.74
distance priors (3)	21.91	15.35	−6.56
SH0ES (1)	6.42	12.07	+5.65
total	1624.85	1601.19	−23.66

The DR2 BAO contribution (−12.77) is approximately twice the DR1 BAO contribution (−5.78); the SH0ES residual increases by +5.65 at LockPhi (the lock prefers $H_0 = 69.43$, further from $H_0^{\mathrm{SH0ES}} = 73.04$ than the LCDM best of $H_0 = 70.40$).

A point-by-point breakdown of the BAO contribution shows four measurements dominate. Their summed $|\Delta\chi^2|$ is approximately 15.5, or 75% of the total BAO improvement of −20.66:

point	z	$\chi^2_{\Lambda\mathrm{CDM}}$	$\chi^2_{\mathrm{LockPhi}}$	$\Delta\chi^2$	note
DR2 LRG2	0.706	7.30	0.10	−7.21	near peak
DR1 LRG1	0.510	7.21	3.11	−4.10	near peak
DR2 LRG1	0.510	5.10	2.69	−2.40	near peak
DR1 LRG2	0.706	4.31	2.55	−1.76	near peak

We additionally run an A-grid scan with LRG1 and LRG2 removed simultaneously from both releases (4 measurements dropped, 8 BAO points removed in total). $(H_0, \Omega_m, \Omega_b h^2)$ are profiled at each A on the LCDM-extended grid:

A	χ^2	$\Delta\chi^2$ vs ΛCDM
0.000 (ΛCDM)	1599.99	0
+0.020	1594.29	−5.70
+0.040	**1592.26**	**−7.73**
+0.060	1593.64	−6.35
+0.080	1598.18	−1.81
−0.020	1609.65	+9.66
−0.050	1632.27	+32.28

With LRG1 and LRG2 removed from both releases, the best-fit amplitude is $A = +0.040$, a -23% shift from the unrestricted value $+0.0520$. The minimum of the A-profile is not at $A = 0$.

7 DR3 prior predictions

Two BAO measurements at $z = 0.510$ (LRG1) and $z = 0.706$ (LRG2) supply approximately 75% of the LockPhi combined BAO improvement (§6.5). The LockPhi vs ΛCDM differences in predicted D_M/r_d and D_H/r_d at these redshifts are listed below, evaluated at each model's combined-fit best:

z	D_M/r_d (ΛCDM)	D_M/r_d (LockPhi)	Δ%	D_H/r_d (ΛCDM)	D_H/r_d (LockPhi)	Δ%
0.510	13.232	13.321	+0.67	22.598	22.012	−2.60
0.706	17.422	17.354	−0.39	20.203	19.543	−3.27

The DR2 (current) measurement uncertainties are $\sigma(D_M/r_d) \approx 0.169$ at $z = 0.510$ and 0.180 at $z = 0.706$. With DR3 expected to reduce uncertainties by approximately a factor of two (5-year DESI vs 1-year), the LockPhi vs ΛCDM difference at each redshift is approximately a 1σ effect at DR3 precision per individual measurement.

The 5×5 landscape of §5.3 implies that a DR3 free-Gaussian fit, if LockPhi describes a real signal at the location identified by the present combined fit, should converge to $z_c \in [0.587, 0.649]$ and $w \in [0.147, 0.199]$ (the range of the $\Delta\text{BIC} < -14$ region). A definitive resolution of the natural-constant identification of z_c would require DR3 free-fit precision better than ±1% in z_c at the LockPhi center, which is at the upper end of expected DR3 capability for a single fit.

8 Status

candidate	status as of v16
v6 dual-field	retired (v7)
v7 wavelet (z_c free)	retired (v8.5)
v9 R6 ($w = 2$)	retired (v10)
v10 R7 ($n = 0$ family)	not directly reverified
v12/v13 C1 ($n = 1$ family)	retired (v15)
v14–note D1 ($z_c = \sqrt{\pi}$)	retired (v15)
LockB ($z_c = 1/\sqrt{\pi}$, $w = 2/(5\pi)$)	live, not closed (status from v15 unchanged)
LockC ($z_c = 4/7$, $w = 2/(5\pi)$)	live, not closed (status from v15 unchanged)
LockPhi ($z_c = 1/\varphi$, $w \in \{e/(5\pi), \pi/18\}$)	**new candidate (v16, this work); not closed**

What v16 establishes that was not established in v15:

1. DR1 BAO and DR2 BAO placed in the same likelihood return a free-Gaussian best-fit at $(z_c, w) = (0.6185, 0.172)$. This position is not equal to either single-release free best.
2. The natural-constant $1/\varphi$ matches the combined free-fit z_c to 0.075%. The matches for w ($e/(5\pi)$, $\pi/18$) are within 1.5% but the present data do not select between these two forms.
3. LockPhi $\Delta\text{BIC} = -16.20$ on the combined likelihood. LockB and LockC, when refit on the same combined likelihood, give -12.22 and -13.27.
4. LockPhi K-A multi-LOO on the combined BAO returns 7 WIN, 4 tie, 0 BURN. The DR1 LRG1 single-LOO failure of v15 (LockB/C) does not occur for LockPhi on the combined data.

5. The natural-constant identification of z_c is not resolved by the present data. Ten candidate values within $\pm 1.5\%$ of $1/\varphi$ all lie within $\Delta\chi^2 < 0.14$ of the minimum; the second-best is the round number 0.6200 at $\Delta\chi^2 = +0.0004$. The LockPhi center sits on a wide ridge in (z_c, w).

The two v15 candidates LockB and LockC are not affected by this work; their v15 status (live, not closed) is preserved. We do not introduce any new family hypothesis. The search for additional candidates is not closed and continues. The next decisive test is DESI DR3.

9 Limitations

1. LockPhi was identified by inspection of the combined free-fit best-fit. By working principle (e), this does not save a parameter; we count $k = 4$ for LockPhi.
2. The natural-constant identification of z_c is not data-resolved (§5.2). The lock at $1/\varphi$ is one of several near-equivalent candidates within the data's resolving power.
3. The two natural-constant forms of w (LockPhi-A, LockPhi-B) are not distinguished by the present data ($\Delta\chi^2 = 0.002$).
4. The four BAO measurements at $(z, \text{release}) = (0.510, \text{DR1}), (0.510, \text{DR2}), (0.706, \text{DR1}), (0.706, \text{DR2})$ supply approximately 75% of the LockPhi BAO improvement. With LRG1 and LRG2 removed from both releases simultaneously, the K-A verdict reaches tie at $\Delta\text{BIC} = -0.28$.
5. The DR1 BAO and DR2 BAO are treated as independent measurements; correlations between them (if present beyond the published covariances) are not modeled.
6. No full Boltzmann Planck likelihood is used. The compressed distance prior calibration of v14 is carried forward.
7. No Lagrangian formulation. The model name LVC remains aspirational.
8. DR3 BAO precision (§7) is the primary discriminating data not yet available.

10 Reproducibility

The numerical results in this paper are produced by an extension of the v15 reproduction script. The combined likelihood is constructed by adding the DR1 BAO block as an additional summand alongside the DR2 BAO block, with BOSS, eBOSS, Pantheon+, the distance priors and SH0ES included once each. The K-A multi-LOO sweep removes labels simultaneously from both BAO releases when the same label appears in both. Tabulated χ^2 values reproduce within ≤ 0.05 and ΔBIC values within ≤ 0.10. A self-contained reproduction script for v16 is in preparation and will be released separately.

Note on scope. This v16 paper is a working document. It identifies one new candidate (LockPhi) on the combined DR1+DR2 BAO likelihood, reports the outcome of direct kill tests applied to it, and records the natural-constant identification of (z_c, w) as a structural fact about where the data points rather than a parameter saving. It does not retire LockB or LockC, does not introduce a new family hypothesis, and does not derive an action. The candidate set as of v16 is LockB, LockC, and LockPhi, all live and not closed. Further candidate search is not closed.

A Effective dark energy at LockPhi

For information rather than evaluation, we record the effective dark-energy density and equation-of-state extracted from $H^2(z) = H^2_{\Lambda\mathrm{CDM}}(z) \cdot T^2(z)$ via

$$\Omega_{\mathrm{DE,eff}}(z) = E^2(z) - \Omega_m(1+z)^3 - \Omega_r(1+z)^4, \qquad w_{\mathrm{eff}}(z) = -1 - \frac{1}{3}\frac{d\ln\rho_{\mathrm{DE,eff}}}{d\ln(1+z)}. \tag{5}$$

For LockPhi at the combined-fit best (using $w = e/(5\pi)$ form, $A = +0.0520$, $z_c = 1/\varphi$): the maximum deviation $|w_{\mathrm{eff}} + 1|$ over $z \in [0, 5]$ is 0.728 at $z = 0.770$ (quintessence side, $w_{\mathrm{eff}} = -0.272$); the deepest phantom value is $w_{\mathrm{eff}} = -1.623$ at $z = 0.508$. The total swing peak-to-peak is 1.351. The $\Omega_{\mathrm{DE,eff}}$ varies by 29.1% relative to its $z = 0$ value over the same range. For comparison, the same quantities at the C1 (DR2) combined fit give swing 1.07, variation 26.7%; at LockB (combined) 1.86 and 31.1%; at LockC (combined) 1.96 and 32.7%. The LockPhi swing is intermediate; its swing per unit $|\Delta\mathrm{BIC}|$ is $1.351/16.20 = 0.083$, the smallest among the four candidates listed.

These quantities are recorded for downstream derivation work and are not used in the kill tests of §6.